



Hunters Hill
High School

Student Number

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2023 TRIAL EXAMINATION

Mathematics Extension 2

General

Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/ or calculations

Total marks:

100

Section I – 10 marks (pages 3–7)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8–13)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 Marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. Let z be a complex number such that $z^2 = \frac{i}{z}$.
Which of the following is a possible value for z ?
- (A) $0 + i$
- (B) $-1 + i$
- (C) $\frac{\sqrt{3}}{2} + \frac{1}{2}i$
- (D) $\frac{\sqrt{3}}{2} - \frac{1}{2}i$
2. Consider the statement: If all walls of this room are green, then this is a green room.
Which of the following statements is the contrapositive of this statement?
- (A) If this is a green room, then all walls of this room are green
- (B) If this is not a green room, then all walls of this room are not green
- (C) If this is not a green room, then no walls of this room are green
- (D) If this is a not green room, then there is at least one wall of this room which is not green

3. Simplify $\left(\frac{1}{2}(\sqrt{3} - i)\right)^{2023}$

(A) $\frac{1}{2}(-\sqrt{3} - i)$

(B) $\frac{1}{2}(-\sqrt{3} + i)$

(C) $\frac{1}{2}(\sqrt{3} - i)$

(D) $\frac{1}{2}(\sqrt{3} + i)$

4. Consider the vector $\vec{u} = 3\vec{i} - \vec{j} - 2\vec{k}$.

Which of the following vectors is perpendicular to $\vec{u}$?

(A) $4\vec{i} + 4\vec{j} + 2\vec{k}$

(B) $3\vec{i} - 1\vec{j} + 5\vec{k}$

(C) $2\vec{i} - 3\vec{j} + 3\vec{k}$

(D) $\vec{i} - 5\vec{j} + 3\vec{k}$

5. Which of the following is equivalent to the expression $\frac{3x + 7}{(x + 1)(x^2 - 2x - 3)}$?

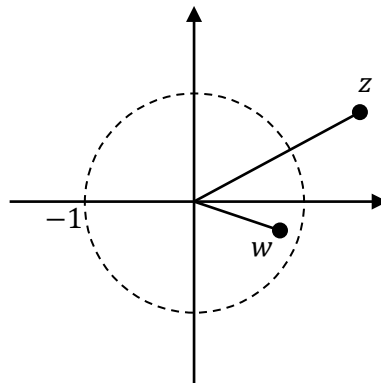
(A) $\frac{-1}{x + 1} + \frac{1}{x - 3}$

(B) $\frac{-1}{(x + 1)^2} + \frac{1}{x - 3}$

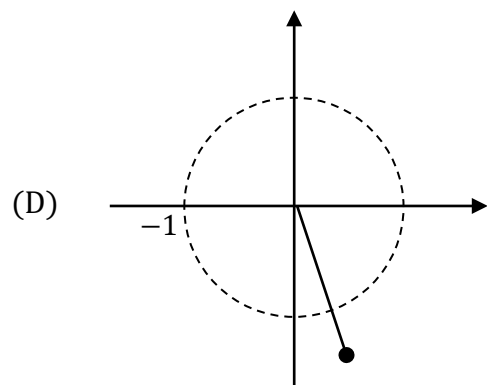
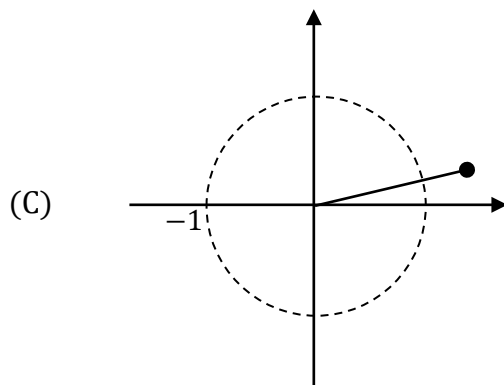
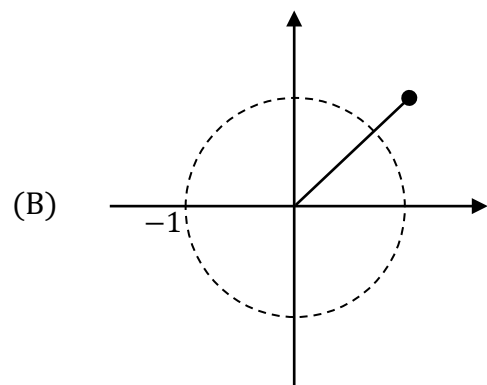
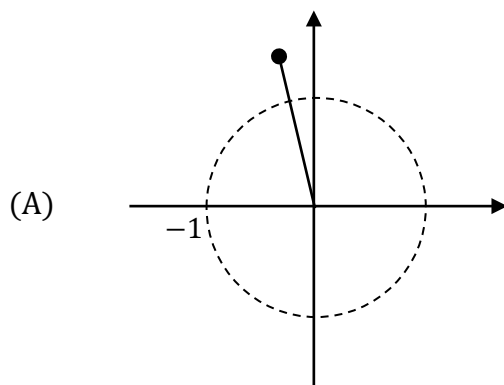
(C) $\frac{-1}{x + 1} + \frac{x - 1}{(x + 1)^2} + \frac{1}{x - 3}$

(D) $\frac{-1}{x + 1} + \frac{-1}{(x + 1)^2} + \frac{1}{x - 3}$

6. The argand diagram shows the complex numbers z and w , where z lies in the first quadrant and w lies in the fourth quadrant.



Which of the following diagrams best illustrates the complex number iwz ?



7. Which of the following expressions is equivalent to $\int -\frac{1}{\sqrt{1-3x^2}} dx$?

- (A) $\frac{1}{6} \cos^{-1} 3x + c$
- (B) $\frac{1}{6} \sin^{-1} 3x + c$
- (C) $\frac{1}{\sqrt{3}} \cos^{-1} \sqrt{3}x + c$
- (D) $\frac{1}{\sqrt{3}} \sin^{-1} \sqrt{3}x + c$

8. A particle is describing simple harmonic motion in a straight line with an amplitude of 6m. Its speed is 4m/s when the particle is 3m from the centre of motion. What is the period of the motion?

- (A) $\frac{3\sqrt{3}\pi}{2}$
- (B) $\frac{9\pi}{2}$
- (C) $\frac{3\sqrt{3}\pi}{4}$
- (D) $\frac{9\pi}{4}$

9. Which of the following represents the vector projection of $\overrightarrow{OA}$ onto $\overrightarrow{OB}$ given $A(1, 3, -2)$ and $B(2, -2, 1)$?

(A) $\begin{bmatrix} 4 \\ -\frac{4}{3} \\ \frac{4}{3} \\ 2 \\ -\frac{2}{3} \end{bmatrix}$

(B) $\begin{bmatrix} 4 \\ \frac{4}{3} \\ \frac{4}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}$

(C) $\begin{bmatrix} 4 \\ -\frac{4}{9} \\ \frac{4}{9} \\ 2 \\ -\frac{2}{9} \end{bmatrix}$

(D) $\begin{bmatrix} -2 \\ \frac{2}{3} \\ -6 \\ \frac{2}{3} \\ \frac{4}{3} \end{bmatrix}$

10. A particle is moving along a straight line so that initially its displacement is $x = 2$, its velocity is $v = 2$ and its acceleration is $a = 4$.

Which of the following is a possible equation of the motion of the particle?

(A) $v = x^2 + 2x$

(B) $v = 2 + 4 \ln(x - 1)$

(C) $v = 2 \sin(x - 2) + 2$

(D) $v^2 = 4(x^2 - 3)$

End of Section I

Section II

90 Marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Begin each question in a new writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Begin a new Writing Booklet.

- (a) Let $z = 1 - 4i$ and $w = 3 + 2i$.
Find:

- | | | |
|-------|---------------------|---|
| (i) | $\bar{w}$ | 1 |
| (ii) | $ wz $ | 2 |
| (iii) | $\frac{\bar{w}}{z}$ | 2 |

- (b) Find

- | | | |
|------|---------------------------------------|---|
| (i) | $\int \frac{dx}{\sqrt{5 + 2x - x^2}}$ | 2 |
| (ii) | $\int \frac{x^2 - 1}{x^2 + 4} dx$ | 2 |

- (c) Solve $z^2 + (2 + i)z + 2 - 2i = 0$ 3

- (d) Prove by contradiction that $\sqrt{7}$ is irrational. 3

End of Question 11

Question 12 (14 marks) Begin a new Writing Booklet.

- (a) Using substitution of $x = \cos^2 \theta$, evaluate 4

$$\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} dx$$

- (b) On an Argand diagram, sketch the region defined by 3

$$-\frac{\pi}{4} < \arg(z - i) < \frac{\pi}{4}$$
$$\operatorname{Re}(z) \leq 3$$

- (c) Find $\int e^x \sin x dx$. 4

- (d) The point $P(x, y)$, representing the complex number z , moves in an Argand diagram such that $|z - 6| = |z + 2i|$.

- (i) Show that the path that P traces out has the formula $3x + y - 8 = 0$. 2

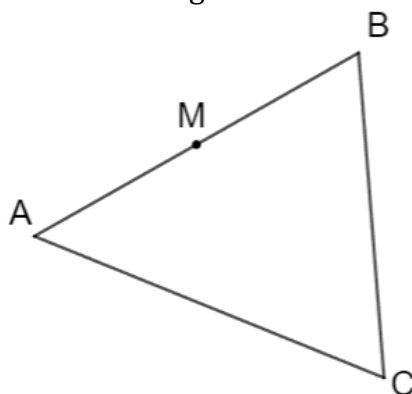
- (ii) Find the minimum value of $|z|$ as P moves along its path. 1

End of Question 12

Question 13 (15 marks) Begin a new Writing Booklet.

- (a) M is the midpoint of side AB in triangle ABC .

4



Using vectors, show that $|\vec{AC}|^2 + |\vec{BC}|^2 = 2(|\vec{CM}|^2 + |\vec{AM}|^2)$.

- (b) A sequence is defined recursively as $u_0 = 5, u_n = 2u_{n-1} + 1$ for $n \geq 1$.

3

By induction, prove that $u_n = 3(2^{n+1}) - 1$ for all positive integers n .

- (c) (i) Express $4i$ in exponential form.

1

- (ii) Hence, find the 4th roots of $4i$.

2

- (iii) Sketch the roots on an Argand diagram.

2

- (d) (i) Prove that $a^2 + b^2 \geq 2ab$

1

- (ii) Hence, or otherwise, prove that $(p+2)(q+2)(p+q) \geq 16pq$, where p and q are positive real numbers.

2

End of Question 13

Question 14 (15 marks) Begin a new Writing Booklet.

- (a) A stone is released on the surface of the ocean, at which point it immediately sinks. Let gravity be 10 ms^{-2} and the resistance due to the water is proportional to the square of the velocity.

- (i) Explain why the acceleration of the stone can be given by 1

$$a = 10 - \frac{k}{m}v^2.$$

- (ii) Given $\frac{k}{m} = \frac{1}{40}$, show that $v = \frac{20(e^t - 1)}{e^t + 1}$. 4

- (iii) Using $a = v \frac{dv}{dx}$, show that $x = 20 \ln \left(\frac{400}{400 - v^2} \right)$. 2

- (iv) How far does the stone sink in the first 3 seconds? 2

- (b) $\tilde{r}_1$ and $\tilde{r}_2$ are two lines with vector equations:

$$\begin{aligned}\tilde{r}_1 &= \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \\ \tilde{r}_2 &= \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} \\ &\text{where } \lambda, \mu \in \mathbb{R}.\end{aligned}$$

- (i) Show that these two lines intersect. 2

- (ii) Find the angle between the lines. 1

- (iii) Find the shortest distance from the point $P(1, 2, 2)$ to the line 3

$$\tilde{r}_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

End of Question 14

Question 15 (16 marks) Begin a new Writing Booklet.

(a) Prove that $9^{n+1} - 2^{n+1}$ is divisible by 7 for all $n \geq 0$. 3

(b) A particle moves in a straight line, experiencing simple harmonic motion. At time t seconds its displacement from a fixed point O in the line is x metres, given by:

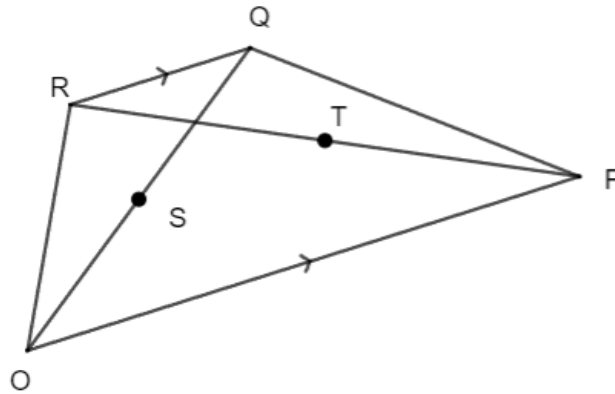
$$x = 1 + \sqrt{2} \cos\left(t - \frac{\pi}{4}\right)$$

(i) Show that $\ddot{x} = -(x - 1)$ 1

(ii) Find the time taken for the particle to first pass through the point O . 2

(iii) Find in simplest exact form, the average speed of the particle during one complete oscillation of its motion. 2

(c) $OPQR$ is a trapezium with $\overrightarrow{OP} = k \overrightarrow{RQ}$.
Let $\underline{p} = \overrightarrow{OP}$ and $\underline{r} = \overrightarrow{OR}$.
 S and T are midpoints of OQ and PR respectively.



(i) Find $\overrightarrow{OS}$ in terms of $\underline{p}$ and $\underline{r}$. 2

(ii) Find $\overrightarrow{ST}$ in terms of $\underline{p}$ and $\underline{r}$. 4

(iii) Deduce the value of k required for $RSTQ$ to be a parallelogram. 2

End of Question 15

Question 16 (15 marks) Begin a new Writing Booklet.

- (a) Evaluate $\int \cos^3 x \sin^2 x \, dx$. 3
- (b) A stranded sailor, armed with a flare stands atop a headland, 40m above sea level. The flare is fired with a velocity of $20\tilde{i} - 6\tilde{j} + 12\tilde{k}$, where $\tilde{i}$, $\tilde{j}$ and $\tilde{k}$ are unit vectors in the east, north and vertically up directions, respectively. The acceleration of the cannonball due to the combined effects of gravity, air resistance and the wind is $-6\tilde{i} + 10\tilde{j} - 4\tilde{k}$.
- (i) Find the displacement vector for the flare with respect to t . 3
- (ii) Find the highest point on the trajectory of the flare. 2
- (iii) Find the time taken for the flare to plunge into the ocean. 1
- (iv) At what angle to the horizontal was the flare fired? 1
- (c) Let $I_n = \int_0^1 (1 - x^2)^{\frac{n}{2}} dx$, where $n \geq 0$ is an integer.
- (i) Show that 3
- $$I_n = \frac{n}{n+1} I_{n-2} \text{ for every integer } n \geq 2.$$
- (ii) Hence evaluate I_5 . 2

End of Examination

2023 12MXX Trial – Solutions and Marking Guidelines

Tuesday, 18 July 2023 6:05 PM

SECTION I

1.

$$z^2 = \frac{i}{z}$$

$$z^3 = i$$

$$(0+i)^3 = -i$$

$$(-1+i)^3 = (\sqrt{2}\text{cis}\frac{3\pi}{4})^3 \neq i$$

$$\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^3 = \left(\text{cis}\frac{\pi}{6}\right)^3$$

$$= \text{cis}\frac{\pi}{2} = i$$

C.

2. $P \Rightarrow Q \rightarrow \neg Q \Rightarrow \neg P$

for all $\rightarrow$ there exists

D

3. $\left(\frac{\sqrt{3}-i}{2}\right)^{2023} = \left(\left(\frac{\sqrt{3}-i}{2}\right)^3\right)^{674} \left(\frac{\sqrt{3}-i}{2}\right)$

$$= i^{674} \cdot \left(\frac{\sqrt{3}-i}{2}\right)$$

$$= i^{672} \cdot i^2 \cdot \left(\frac{\sqrt{3}-i}{2}\right)$$

$$= (+1)(-1) \left(\frac{\sqrt{3}-i}{2}\right)$$

$$= -\frac{\sqrt{3}}{2} + \frac{i}{2}$$

$\left. \begin{matrix} i \\ -1 \\ -i \\ +1 \end{matrix} \right\}$

B

4. $\underline{u} = 3\underline{i} - \underline{j} - 2\underline{k}$

A: $4(3) + 4(-1) + 2(2) = 4$

B: $3(3) - 1(-1) + 5(-2) = 0$

C: $2(3) - 3(1) + 3(-2) = 3$

D: $1(3) - 5(1) + 3(-2) = 2$

B.

5. $\frac{3x+7}{(x+1)^2(x-3)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-3}$

$$3x+7 = A(x+1)(x-3) + B(x-3) + C(x+1)^2$$

$$x=-1 \Rightarrow 4 = -4B$$

$$B = -1$$

$$x=3 \Rightarrow 16 = 4^2C$$

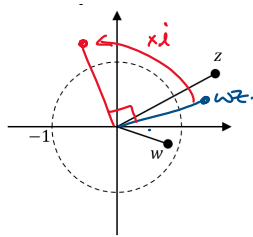
$$C = 1$$

$$x^2: \Rightarrow 0 = A + C$$

$$A = -1$$

D

6. product of $wz \Rightarrow$ add arguments
 $\Rightarrow |w| < 1 \wedge$



$$|wz| < |z|$$

wz is a rotation
 $\frac{\pi}{2}$ anticlockwise

A

7. $\frac{1}{\sqrt{3}} \int -\frac{\sqrt{3} dx}{\sqrt{1-(\sqrt{3}x)^2}} = \frac{1}{\sqrt{3}} \cos^{-1} \sqrt{3}x dx$

C

8.



$$v^2 = n^2 (a^2 - x^2)$$

$$4^2 = n^2 (6^2 - 3^2)$$

$$16 = n^2 (36 - 9)$$

$$n^2 = \frac{16}{27}$$

$$n = \frac{4}{3\sqrt{3}} \Rightarrow P = \frac{2\pi \times \frac{4}{3\sqrt{3}}}{\frac{1}{A_2}} = \frac{3\sqrt{3}\pi}{2}$$

A

9.

$$\frac{a \cdot b}{b \cdot b} \cdot b = \frac{1(2) + 3(-2) - 2(1)}{2^2 + (-2)^2 + 1^2} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$$= \frac{-6}{9} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$$= -\frac{2}{3} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

A

10.

$x=2, y=2$ doesn't satisfy A or D

$$y = 2\sin(x-2) + 2$$

$$a = (2\sin(x-2) + 2) \cdot 2\cos(x-2) \quad \left(\frac{y dy}{dx}\right)$$

$$= (2\sin 0 + 2) 2\cos 0$$

$$= 4$$

C

Question 11

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a. $z = 1-4i$ $w = 3+2i$
i. $\bar{w} = 3-2i$

ii. $|wz| = |(1-4i)(3+2i)|$
 $= |11-10i|$
 $= \sqrt{11^2+10^2}$
 $= \sqrt{221}$

iii. $\frac{\bar{w}}{z} = \frac{3-2i}{1-4i} \times \frac{1+4i}{1+4i}$
 $= \frac{3-2i+12i+8}{1+4^2}$
 $= \frac{11+10i}{17}$

b. i. $\int \frac{dx}{\sqrt{5-2x-x^2}} = \int \frac{dx}{\sqrt{6-(x^2-2x+1)}}$
 $= \int \frac{dx}{\sqrt{6-(x-1)^2}}$
 $= \frac{\sin^{-1} \frac{x-1}{\sqrt{6}}}{\sqrt{6}} + c$

ii. $\int \frac{x^2-1}{x^2+4} dx = \int \frac{x^2+4-5}{x^2+4} dx$
 $= \int \left(1 - \frac{5}{x^2+4}\right) dx$
 $= x - \frac{5}{2} \tan^{-1} \frac{x}{2} + c$

c. $z^2 + (2+i)z + 2-2i = 0$
 $z = \frac{-(2+i) \pm \sqrt{(2+i)^2 - 4(1)(2-2i)}}{2(1)}$

$\Delta = 3+4i-8+4i$
 $= -5+12i$
 $\text{let } (x+iy)^2 = -5+12i$
 $\text{Re+Im } x^2-y^2 = -5$
 $2xy = 12 \Rightarrow xy = 6$

by observation $x=2, y=3$

$\therefore \sqrt{-5+12i} = \pm(2+3i)$
 $\therefore z = \frac{-2-i \pm (2+3i)}{2}$
 $= \frac{2i}{2}, \frac{-4-4i}{2}$
 $= i, -2-2i$

1- correct answer

1- correct product of wz or magnitude after error

2- correct answer

1- multiplies by conjugate

2- correct answer

1- manipulates to find perfect square

2- correct ans. (ignore +c)

1- correctly separates fraction

2- correct ans (ignore c)

1- correct use of quadratic formula

2- solution for x and y.

3- correct ans

d. let $\sqrt{7}$ be rational

$$\therefore \sqrt{7} = \frac{a}{b} \quad a, b \in \mathbb{R}$$

and a and b have no common divisor other than 1.

$$a = b\sqrt{7}$$

$$a^2 = 7b^2$$

as 7 divides a^2 , then 7 must divide a .

hence $a = 7c$ for constant c .

$$\therefore (7c)^2 = 7b^2$$

$$\text{and } 7c^2 = b^2$$

as 7 divides b^2 , then 7 must divide b

$\therefore$ both a and b have a factor of 7, contradicting the earlier assumption

$\therefore$ by contradiction, $\sqrt{7}$ is irrational

1 - assumes $\sqrt{7}$ is rational
in form $\frac{a}{b}$

2 - deduces that 7 divides a , or equivalent.

3 - complete proof.

Question 12

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a) $x = \cos^2 \theta$
 $dx = 2 \cos \theta \cdot (-\sin \theta) d\theta$

for $x=0$, $\theta = \frac{\pi}{2}$
 $x = \frac{1}{2}$, $\theta = \frac{\pi}{4}$

$$\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} dx = -2 \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \sqrt{\frac{\cos^2 \theta}{1-\cos^2 \theta}} \cdot \cos \theta \sin \theta d\theta$$

$$= 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sqrt{\frac{\cos^2 \theta}{\sin^2 \theta}} \cdot \cos \theta \sin \theta d\theta$$

$$= 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos \theta \cdot \cos \theta \sin \theta}{\sin \theta} d\theta$$

$$= 2 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$= \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta$$

$$= \left[\theta + \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} + \frac{1}{2} \sin \pi - \left(\frac{\pi}{4} + \frac{1}{2} \sin \frac{\pi}{2} \right)$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

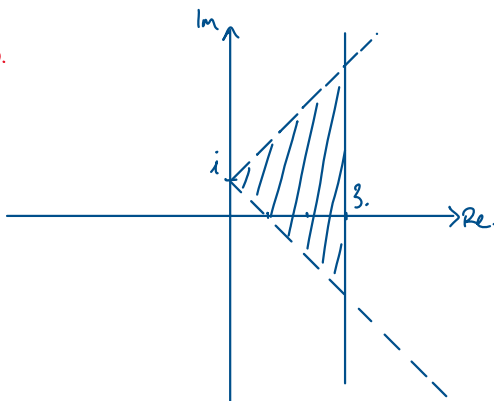
1- correct dx

2- correct substitution, including bounds

3- correct simplification

4- ans

b.



1- any correct line
 $\arg(z-i) = \pm \frac{\pi}{4}$
 $\operatorname{Re}(z) = 3$

2- a correct region

3- ans

c. $\int e^x \sin x dx$

let $u = \sin x$ $dv = e^x dx$

$du = \cos x dx$ $v = e^x$

$$\int e^x \sin x dx = e^x \sin x - \int e^x \cos x dx$$

let $u = \cos x$ $dv = e^x dx$

$du = -\sin x dx$ $v = e^x$

$$= e^x \sin x - \left(e^x \cos x - \int e^x (-\sin x) dx \right)$$

$$= e^x \sin x - \cos x - \int e^x \sin x dx$$

1- correct du and v.

2- correct use of by parts

3- second use of by parts

$$2 \int e^x \sin x dx = e^x (\sin x - \cos x)$$

$$\therefore \int e^x \sin x dx = \frac{e^x}{2} (\sin x - \cos x)$$

d. i. $|z-6| = |z+2i|$

$$|x+iy-6| = |x+iy+2i|$$

$$|x-6+iy|^2 = |x+i(y+2)|^2$$

$$(x-6)^2 + y^2 = x^2 + (y+2)^2$$

$$\cancel{x^2} - 12x + 36 + \cancel{y^2} = \cancel{x^2} + \cancel{y^2} + 4y + 4$$

$$12x + 4y - 32 = 0$$

$$\therefore 3x + y - 8 = 0$$

ii. minimum value occurs when z is nearest the origin $\Rightarrow$ distance from $(0,0)$ to $3x+y-8=0$

$$d = \frac{|3(0)+0-8|}{\sqrt{3^2+1^2}}$$

$$= \frac{|-8|}{\sqrt{10}}$$

$$= \frac{4\sqrt{10}}{5}$$

4 - ans

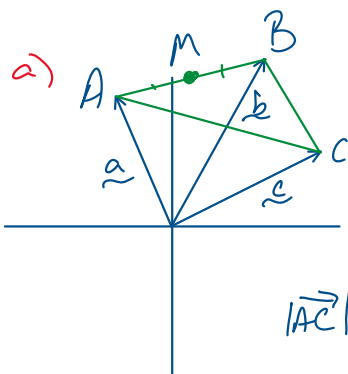
1 - correct magnitude expression for one side

2 - ans

1 - correct ans.

Question 13

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$$\begin{aligned}\vec{OM} &= \vec{a} + \frac{1}{2}(\vec{b} - \vec{a}) \\ &= \frac{1}{2}(\vec{a} + \vec{b})\end{aligned}$$

$$\begin{aligned}|\vec{AC}|^2 + |\vec{BC}|^2 &= (\vec{c} - \vec{a}) \cdot (\vec{c} - \vec{a}) + (\vec{c} - \vec{b}) \cdot (\vec{c} - \vec{b}) \\ &= \vec{c}^2 - 2\vec{a} \cdot \vec{c} + \vec{a}^2 + \vec{c}^2 - 2\vec{b} \cdot \vec{c} + \vec{b}^2\end{aligned}$$

$$\begin{aligned}|\vec{CM}|^2 + |\vec{AM}|^2 &= \left(\frac{1}{2}(\vec{a} + \vec{b}) - \vec{c}\right) \cdot \left(\frac{1}{2}(\vec{a} + \vec{b}) - \vec{c}\right) + \left(\frac{1}{2}(\vec{a} + \vec{b}) - \vec{a}\right) \cdot \left(\frac{1}{2}(\vec{a} + \vec{b}) - \vec{a}\right) \\ &= \frac{1}{4}(\vec{a} + \vec{b})^2 - (\vec{a} + \vec{b}) \cdot \vec{c} + \vec{c}^2 + \frac{1}{4}(\vec{a} + \vec{b})^2 - \vec{a} \cdot (\vec{a} + \vec{b}) + \vec{a}^2 \\ &= \frac{1}{2}(\vec{a} + \vec{b})^2 - \vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c} + \vec{c}^2 - \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} + \vec{a}^2 \\ &= \frac{1}{2}\vec{a}^2 + \vec{a} \cdot \vec{b} + \frac{1}{2}\vec{b}^2 - \vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c} + \vec{c}^2 - \vec{a} \cdot \vec{b} \\ &= \frac{1}{2}(\vec{a}^2 + \vec{b}^2 - 2\vec{a} \cdot \vec{c} - 2\vec{b} \cdot \vec{c} + 2\vec{c}^2)\end{aligned}$$

$$\therefore |\vec{AC}|^2 + |\vec{BC}|^2 = 2(|\vec{CM}|^2 + |\vec{AM}|^2)$$

b) $u_0 = 5, u_n = 2u_{n-1} + 1 \quad n \geq 1$

Prove true for $n=0$

$$\begin{aligned}u_0 &= 3(2^{0+1}) - 1 \\ &= 3 \times 2 - 1 \\ &= 5\end{aligned}$$

$\therefore$ expression true for $n=0$

Assume true for $n=k$

$$\therefore u_k = 3(2^{k+1}) - 1 \quad \text{induction hypothesis}$$

Prove true for $n=k+1$

$$\text{i.e. } u_{k+1} = 3(2^{k+2}) - 1$$

$$\begin{aligned}u_{k+1} &= 2u_k + 1 \\ &= 2 \times (3(2^{k+1}) - 1) + 1 \quad \text{by induction hypothesis} \\ &= 3(2 \times 2^{k+1}) - 2 + 1 \\ &= 3(2^{k+2}) - 1 \quad \text{as required}\end{aligned}$$

$\therefore$ by the process of induction, expression's true for all positive integers n .

1 - finds $\vec{c}$

2 - expands $|\vec{PR}|^2$ or eq. 10

3 - approp. 1 with both

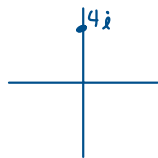
4 - correct $\leq$

1 - proves base

2 - correctly hypothesis

3 - correct pro

c. i. $4i = 4e^{i\frac{\pi}{2}}$



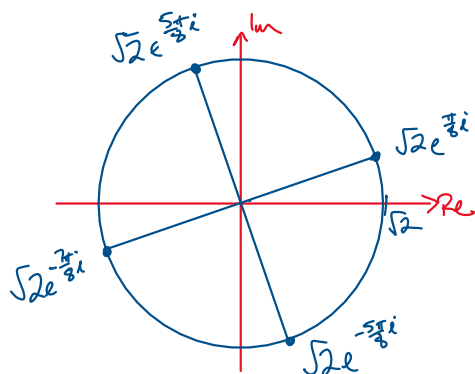
ii. Let $z^4 = 4i$
 $= 4e^{i(\frac{\pi}{2} + 2\pi k)}$

$\therefore z = \sqrt[4]{4} e^{i\frac{\pi}{8}(1+4k)}$

$\therefore z = \sqrt{2} e^{i\frac{\pi}{8}}, \sqrt{2} e^{i\frac{5\pi}{8}}, \sqrt{2} e^{i\frac{9\pi}{8}}, \sqrt{2} e^{i\frac{13\pi}{8}}$
 $= \sqrt{2} e^{i\frac{\pi}{8}}, \sqrt{2} e^{i\frac{5\pi}{8}}, \sqrt{2} e^{-i\frac{7\pi}{8}}, \sqrt{2} e^{-i\frac{3\pi}{8}}$

Correct this to $-3/8$ instead of $-5/8$

iii.



d. i. $(a-b)^2 \geq 0$ for all $a, b > 0$.

$a^2 - 2ab + b^2 \geq 0$
 $\therefore a^2 + b^2 \geq 2ab$

ii. From i) $(\sqrt{p})^2 + (\sqrt{2})^2 \geq 2\sqrt{p} \cdot \sqrt{2}$
 $(\sqrt{q})^2 + (\sqrt{2})^2 \geq 2\sqrt{q} \cdot \sqrt{2}$
 $(\sqrt{p})^2 + (\sqrt{2})^2 \geq 2\sqrt{p} \cdot \sqrt{q}$

$\therefore (p+2)(q+2)(p+q) = 2\sqrt{2}\sqrt{p} \cdot 2\sqrt{2}\sqrt{q} \cdot 2\sqrt{p}\sqrt{q}$
 $= 8pq$

1-correct ans

1-correct form

2-correct re

1- 4 equally sp
on circle
or

2-correct st

1-correct pr

1- notes one a
these

2-correct pr

Question 14

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a) i. Taking downwards as positive direction with resistance acting against it.



$$\text{so } ma = mg - kv^2$$

$$\text{and } a = g - \frac{k}{m}v^2$$

ii. $\frac{k}{m} = \frac{1}{40}$

$$\begin{aligned} \frac{dv}{dt} &= 10 - \frac{1}{40}v^2 \\ &= \frac{1}{40}(400 - v^2) \end{aligned}$$

$$\int_0^v \frac{dv}{400 - v^2} = \int_0^t \frac{1}{40} dt$$

$$\begin{aligned} \text{let } \frac{1}{400 - v^2} &= \frac{A}{20 - v} + \frac{B}{20 + v} \\ 1 &= A(20 + v) + B(20 - v) \\ \Rightarrow A &= B = \frac{1}{40} \end{aligned}$$

$$\therefore \frac{1}{40} \int_0^v \left(\frac{1}{20 - v} + \frac{1}{20 + v} \right) dv = \frac{t}{40}$$

$$\left[-\ln|20 - v| + \ln|20 + v| \right]_0^v = t$$

$$\ln\left(\frac{20 + v}{20 - v}\right) - \ln\left(\frac{20}{20}\right) = t$$

$$\therefore \ln\left(\frac{20 + v}{20 - v}\right) = t$$

$$\frac{20 + v}{20 - v} = e^t$$

$$20 + v = e^t(20 - v)$$

$$\begin{aligned} v(1 + e^t) &= 20(e^t - 1) \\ \therefore v &= 20\left(\frac{e^t - 1}{e^t + 1}\right) \end{aligned}$$

iii. $v \frac{dv}{dx} = \frac{1}{40}(400 - v^2)$

$$\begin{aligned} \frac{1}{2} \int_0^v \frac{2v dv}{400 - v^2} &= \frac{1}{40} \int_0^x dx \\ \frac{1}{2} \left[\ln|400 - v^2| \right]_0^v &= \frac{x}{40} \end{aligned}$$

$$\begin{aligned} x &= -20 \left(\ln|400 - v^2| - \ln(400) \right) \\ &= 20 \left(\ln 400 - \ln(400 - v^2) \right) \end{aligned}$$

$$= 20 \ln\left(\frac{400}{400 - v^2}\right)$$

1 - reasonable explanation noting direction of travel

1 - correct separation

2 - partial fractions used correctly

3 - correct integration

4 - ans shown

1 - correct separation

2 - ans shown

$$\begin{aligned}
 \text{iv } at=3 \quad v &= \frac{20(e^3-1)}{e^3+1} \\
 &= 18.10296507 \text{ ms}^{-1} \\
 x &= 20 \ln \left(\frac{400}{400 - 18.103} \right) \\
 &= 34.2176684 \text{ m}
 \end{aligned}$$

1- correct v

2- correct x

$$\begin{aligned}
 \text{b) i. } \underline{r}_1 &= \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \\
 \underline{r}_2 &= \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{intersect when } 2+0\lambda &= 1+\mu & - \\
 0-\lambda &= 2+3\mu & - \\
 1+\lambda &= 0-4\mu & -
 \end{aligned}$$

$$\begin{aligned}
 \text{from i } \mu &= 1 \\
 \text{from ii } -\lambda &= 2+3(1) \\
 \lambda &= -5
 \end{aligned}$$

$$\begin{aligned}
 \text{given iii} \\
 \text{LHS} &= 1-5 & \text{RHS} &= -4(1) \\
 &= -4 & &= -4 \\
 & & &= \text{LHS}
 \end{aligned}$$

$\therefore$ the lines intersect.

1- finds correct μ

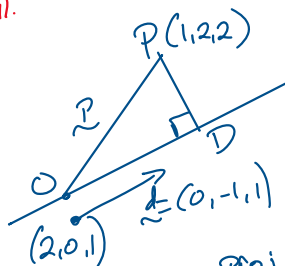
2- confirms intersection

$$\text{ii. let direction vectors } \underline{d}_1 = (0, -1, 1) \text{ and } \underline{d}_2 = (1, 3, -4)$$

$$\begin{aligned}
 \cos \theta &= \frac{\underline{d}_1 \cdot \underline{d}_2}{|\underline{d}_1| |\underline{d}_2|} \\
 &= \frac{0(1) - 1(3) + 1(-4)}{\sqrt{0^2 + (-1)^2 + 1^2} \sqrt{1^2 + 3^2 + (-4)^2}} \\
 &= \frac{-7}{\sqrt{2} \cdot \sqrt{26}} \\
 &= \frac{-7}{2\sqrt{13}} \\
 \theta &= 166.1^\circ
 \end{aligned}$$

1- correct angle.

iii.



$$\underline{r} = (-1, 2, 1)$$

$$\begin{aligned}
 \text{proj}_{\underline{d}} \underline{r} &= \frac{(0, -1, 1) \cdot (-1, 2, 1)}{(0, -1, 1) \cdot (0, -1, 1)} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \\
 &= \frac{-2+1}{1+1} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \\
 &= \frac{1}{2} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}
 \end{aligned}$$

1- finds projection

$$\begin{aligned}
 \vec{DP} &= \vec{P} - \text{proj}_{\vec{d}} \vec{P} \\
 &= (-1, 2, 1) - \left(0, \frac{1}{2}, -\frac{1}{2}\right) \\
 &= \left(-1, \frac{3}{2}, \frac{3}{2}\right)
 \end{aligned}$$

2- finds perpendicular vector

$$\begin{aligned}
 |\vec{DP}| &= \sqrt{(-1)^2 + \left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^2} \\
 &= \sqrt{\frac{11}{2}} \\
 &= \frac{\sqrt{22}}{2} \text{ units}
 \end{aligned}$$

3- finds distance.

Question 15

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a. Prove base case, $n=0$

$$9^{0+1} - 2^{0+1} = 9 - 2 = 7$$

which is divisible by 7,
hence true for $n=0$.

1 - proves base case true,

Assume true for $n=k$

$$\therefore 9^{k+1} - 2^{k+1} = 7M, \quad M \in \mathbb{Z}.$$

and $9^{k+1} = 7M + 2^{k+1}$ - induction hypothesis

Prove true for $n=k+1$

$$\begin{aligned} 9^{(k+1)+1} - 2^{(k+1)+1} &= 9 \cdot 9^{k+1} + 2 \cdot 2^{k+1} \\ &= 9(7M + 2^{k+1}) + 2 \cdot 2^{k+1} \quad \text{by induction hypothesis} \\ &= 7(9M) + 9 \cdot 2^{k+1} + 2 \cdot 2^{k+1} \\ &= 7(9M) + 7 \cdot 2^{k+1} \\ &= 7(9M + 2^{k+1}) \end{aligned}$$

2 - correct use of hypothesis
- correct induction step w/ correct base case

which is divisible by 7 as $9M + 2^{k+1} \in \mathbb{Z}$.

3 - correct proof

$\therefore$ by principle of mathematical induction,
expression divisible by 7 for $n \geq 0$.

b. i. $x = 1 + \sqrt{2} \cos(t - \frac{\pi}{4}) \Rightarrow \sqrt{2} \cos(t - \frac{\pi}{4}) = x - 1$

$$\dot{x} = -\sqrt{2} \sin(t - \frac{\pi}{4})$$

$$\ddot{x} = -\sqrt{2} \cos(t - \frac{\pi}{4})$$

$$= -(x - 1)$$

1 - correctly shown

ii. Solve, $1 + \sqrt{2} \cos(t - \frac{\pi}{4}) = 0$

$$\cos(t - \frac{\pi}{4}) = -\frac{1}{\sqrt{2}}$$

$$t - \frac{\pi}{4} = (2n+1)\pi \pm \frac{\pi}{4} \quad n \in \mathbb{Z}.$$

$$\begin{aligned} t &= (2n+1)\pi, (2n+1)\pi + \frac{\pi}{2} \\ &= \pi, \frac{3\pi}{2}, 3\pi, \frac{7\pi}{2}, \dots \end{aligned}$$

$\therefore$ particle first passes through 0 at π seconds

1 - shows expression

2 - correct ans.

iii. distance travelled during one oscillation is $4 \times$ amplitude

$$\text{so } d = 4\sqrt{2}$$

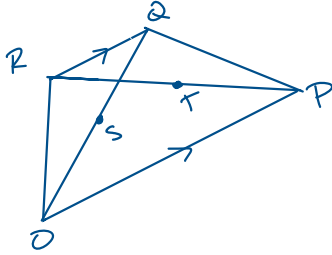
1 - finds distance

$$u = v = 10\pi$$

$$\text{period} = 2\pi$$

$$\begin{aligned} \text{so average speed} &= \frac{4\sqrt{2}}{2\pi} \\ &= \frac{2\sqrt{2}}{\pi} \\ &= 0.900316 \text{ ms}^{-1} \end{aligned}$$

C. i.



$$\begin{aligned} \vec{OS} &= \frac{1}{2}(\vec{OQ}) \\ &= \frac{1}{2}(\vec{OR} + \vec{RQ}) \\ &= \frac{1}{2}(\vec{OR} + \frac{1}{k}\vec{OP}) \\ &= \frac{1}{2}(\vec{r} + \frac{1}{k}\vec{p}) \end{aligned}$$

ii.

$$\begin{aligned} \vec{OT} &= \vec{OR} + \frac{1}{2}\vec{RP} \\ &= \vec{OR} + \frac{1}{2}(\vec{OP} - \vec{OR}) \\ &= \frac{1}{2}(\vec{r} + \vec{p}) \end{aligned}$$

$$\begin{aligned} \vec{ST} &= \vec{OT} - \vec{OS} \\ &= \frac{1}{2}(\vec{r} + \vec{p}) - \frac{1}{2}(\vec{r} + \frac{1}{k}\vec{p}) \\ &= \frac{1}{2}\vec{p}(1 - \frac{1}{k}) \\ &= \frac{k-1}{2k}\vec{p} \end{aligned}$$

iii. for RSTQ to be a parallelogram,

$$\vec{RQ} = \vec{ST}$$

$$\frac{1}{k}\vec{p} = \frac{k-1}{2k}\vec{p}$$

$$\therefore 2 = k-1$$

$$k = 3$$

1 - correct ans

2 - correct ans

1 - $\vec{OS}$ in terms of $\vec{OR}$ and $\vec{RQ}$

2 - ans

1 - in terms of $\vec{OR}$ and $\vec{RP}$

2 - in terms of $\vec{r}$ and $\vec{p}$

3 - correct difference for $\vec{ST}$

4 - ans

1 - equates vectors

2 - correct k.

Question 16

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$$\begin{aligned} a) \quad & \int \cos^3 x \sin^2 x \, dx \\ &= \int (1 - \sin^2 x) \sin^2 x \cdot \cos x \, dx \\ &= \int (\sin^2 x - \sin^4 x) \cos x \, dx \\ &= \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + c \end{aligned}$$

- 1- splits off $\cos x$
- 2- expands $\sin^2$
- 3- integrates

$$\begin{aligned} b) \quad i. \quad & \ddot{x} = -6\hat{i} + 10\hat{j} - 4\hat{k} \\ & \dot{x} = -6t\hat{i} + 10t\hat{j} - 4t\hat{k} + c \end{aligned}$$

$$\text{as } \dot{x} = 20\hat{i} - 6\hat{j} + 12\hat{k} \text{ at } t=0$$

$$\text{then } c = 20\hat{i} - 6\hat{j} + 12\hat{k}$$

$$\text{so } \dot{x} = (20-6t)\hat{i} + (10t-6)\hat{j} + (12-4t)\hat{k}$$

$$x = (20t-3t^2)\hat{i} + (5t^2-6t)\hat{j} + (12t-2t^2)\hat{k} + c$$

$$\text{at } t=0, \quad c = x = 40\hat{k}$$

$$\text{so } x = (20t-3t^2)\hat{i} + (5t^2-6t)\hat{j} + (12t-2t^2+40)\hat{k}$$

- 1- integrates correctly

2- find c .

3- correct ans

$$ii \quad \text{highest point when the vertical component of } \dot{x} \text{ is } 0$$

$$\text{i.e. } 12-4t = 0$$

$$t = 3$$

$$\begin{aligned} x &= (20(3)-3(3)^2)\hat{i} + (5(3)^2-6(3))\hat{j} + (12(3)-2(3)^2+40)\hat{k} \\ &= 33\hat{i} + 27\hat{j} + 58\hat{k} \end{aligned}$$

$$\text{Highest point is } (33, 27, 58)$$

1- correct height 58m

2- correct ans

$$iii \quad \text{flame reaches ocean when vertical component of } x \text{ is } 0$$

$$\therefore 40+12t-2t^2 = 0$$

$$20+6t-t^2 = 0$$

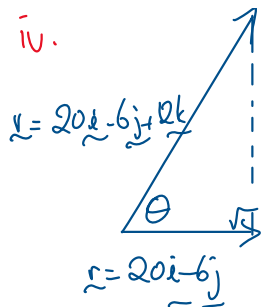
$$20+9-(t^2-6t+9) = 0$$

$$\begin{aligned} (t-3)^2 &= 29 \\ t &= 3 + \sqrt{29} \end{aligned}$$

disregard $t < 0$

$$= 8.3851648 \text{ s}$$

1- correct ans



$$\begin{aligned}\cos\theta &= \frac{\underline{r} \cdot \underline{r}}{|\underline{r}| |\underline{r}|} \\ &= \frac{20^2 + (-6)^2 + 12^2}{\sqrt{20^2 + (-6)^2 + 12^2} \sqrt{20^2 + (-6)^2}} \\ &= \frac{436}{580} \\ &= 0.751724138 \\ \theta &= 29.48580^\circ\end{aligned}$$

1 - correct ans

v)

$$I_n = \int_0^1 (1-x^2)^{\frac{n}{2}} dx$$

$$u = (1-x^2)^{\frac{n}{2}} \quad du = dx$$

$$du = \frac{n}{2}(-2x)(1-x^2)^{\frac{n}{2}-1} dx \quad v = x$$

1 - correct parts

$$\begin{aligned}I_n &= \left[x(1-x^2)^{\frac{n}{2}} \right]_0^1 - \frac{2n}{2} \int_0^1 x^2 (1-x^2)^{\frac{n}{2}-1} dx \\ &= -n \int_0^1 x^2 (1-x^2)^{\frac{n}{2}-1} dx \\ &= -n \int_0^1 \left((1-x^2)^{\frac{n}{2}-1} x^2 (1-x^2)^{\frac{n}{2}-1} - (1-x^2)^{\frac{n}{2}-1} \right) dx \\ &= -n \int_0^1 (1-x^2)(1-x^2)^{\frac{n}{2}-1} dx + n \int_0^1 (1-x^2)^{\frac{n}{2}-1} dx \\ &= -n \int_0^1 (1-x^2)^{\frac{n}{2}} dx + n \int_0^1 (1-x^2)^{\frac{n}{2}-1} dx \\ &= -n I_n + n I_{n-2}\end{aligned}$$

2 - correct substs.

* - appropriate manipulation

$$\therefore (n+1) I_n = n I_{n-2}$$

$$I_n = \frac{n}{n+1} I_{n-2}$$

3 - correctly shown

vi)

$$\begin{aligned}I_1 &= \int_0^1 (1-x^2)^{\frac{1}{2}} dx \\ &= \int_0^{\frac{\pi}{2}} \sqrt{1-\cos^2\theta} \cdot \sin\theta d\theta \quad \begin{array}{l} \text{let } x = \cos\theta \\ dx = -\sin\theta d\theta \end{array} \quad \begin{array}{l} x=1, \theta=0 \\ x=0, \theta=\frac{\pi}{2} \end{array} \\ &= \int_0^{\frac{\pi}{2}} \sin^2\theta d\theta \quad \cos 2\theta = 1 - 2\sin^2\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2\theta) d\theta \\ &= \frac{1}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left(\frac{\pi}{2} + \frac{1}{2} (0) - (0 + \frac{1}{2} (0)) \right) \\ &= \frac{\pi}{4} \\ I_3 &= \frac{3}{3+1} I_1 \\ &= \frac{3}{4} \cdot \frac{\pi}{4}\end{aligned}$$

1 - value for I_1

$$\begin{aligned}
 &= \frac{3\pi}{16} \\
 I_5 &= \frac{5}{5+1} \cdot I_3 \\
 &= \frac{5}{6} \cdot \frac{3\pi}{16} \\
 &= \frac{5\pi}{32}
 \end{aligned}$$

2-ans